

WORD PROBLEMS:

1. Translating English into Mathematics:

- a) Identify the quantity that you have been asked to find and call it x .
- An integer is added... Find the integer. Start with "Let the integer be x ..."
 - A goods train travels... For how many hours does the train travel? Start with "Let the train travel for x hours..."
- b) Make sure that you don't mix the units involved.
- If a problem mentions minutes and hours, work only in hours or only in minutes.
 - Only add like terms. You cannot add money to people and get money.
- c) When you have two expressions in an equation, make sure you know which one is larger.
- d) Draw a table or sketch if possible.
- e) Make sure you know the correct English term for each operation..

ENGLISH	MATHEMATICS
The sum of	+
The difference between	-
The product of	$\times$
The quotient of	$\div$
A number increased by 3	$x + 3$
A number decreased by 3	$x - 3$
Three times a number	$3x$
A number 3 less than x	$x - 3$
A number 3 more than x	$x + 3$
The reciprocal of x	$\frac{1}{x}$
Two consecutive numbers	x and $x + 1$
An even number	$2x$
An odd number	$2x + 1$
Two consecutive even numbers	x and $x + 2$

- f) Once you have solved the equation make sure that you state the solution clearly.

2. PRODUCT PROBLEMS:

a) AGES

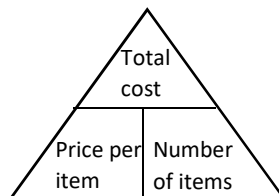
Sara's father is six times as old as Sara. The product of their ages is 150 years. What are their respective ages?

b) NUMBERS

The product of two consecutive integers is 72. Find the integers.

c) SHOPPING

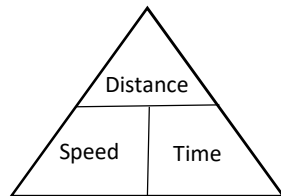
It is useful to draw a table with these types of problems using the triangle below. It is important to note that when you draw the table you have a **variable column** (what you want), a **constant column** (given in the question) and a **formula column**. The formula column is filled in using the relevant formula and information from the other two columns **not** from the question. You then use the **formula column** and any other relevant information given in the question to make an equation.



A shopkeeper orders a number of glasses for which he pays R300. As they are being unpacked, 10 get broken. He sells each glass at R4 more than he paid for it and makes a profit of R100. How many glasses did he originally order and what was the price of each?

d) SPEED, DISTANCE AND TIME

Most speed, distance and time problems are solved using a table in the same way as you have done in shopping problems and using the triangle below.



A man travels 180 km from his farm to town in a loaded truck. On the return journey he is able to travel 30 km/h faster having unloaded his produce in town. He saves 1 hour on the return by travelling faster. At what speed did he travel to town?

e) AREA

A small rectangular vegetable garden is enlarged by increasing the length by 3 m and the width by 1 m. The area of the new garden is three times larger than that of the original garden. Determine the original dimensions of the garden, if its area was 6 m^2 .

3. RATE OF FLOW (RATE OF WORK) PROBLEMS

Rate of flow problems deal with objects or people completing a specific task in a certain time. In general, the problems work with inverse proportion as the more objects or people you add, the less time a task will take to complete.

A tank has two pipes entering it. When operating together, they fill the tank in 40 min. Operating independently, one of the pipes fills the tanks 60 min faster than the other one on its own. Find the time taken by each pipe to fill the tank on its own.